IEOR E4004: Optimization Models and Methods
Your Name:

In Class Exercise

February 2, 2026

Consider the following linear program:

$$\begin{aligned} \min \quad & z = 50x_1 + 100x_2 \\ \text{s.t.} \quad & 7x_1 + 2x_2 \geq 28 \\ & 2x_1 + 12x_2 \geq 24 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Tasks:

1. Rewrite each constraint as a line in the form $x_2 = f(x_1)$.
2. Plot each constraint on the coordinate system below.
3. Clearly indicate the feasible region.
4. Identify all corner points of the feasible region.
5. Compute the objective value at each corner point.
6. Indicate the optimal solution and optimal objective value.

Graph (use this space for your solution):